



Research paper

Ice load sensing of floating nuclear power plant using a reduced mathematical model

Donghui Wang^a, Jiajun Wang^b, Meng Zhang^{a,b,c,*}, Xianqiang Qu^{a,b}^a College of Shipbuilding Engineering, Harbin Engineering University, Harbin, 150001, China^b Yantai Research Institute, Harbin Engineering University, Yantai, 264000, China^c College of Power and Energy Engineering, Harbin Engineering University, Harbin, 150001, China

ARTICLE INFO

Keywords:

Floating nuclear power plant
Reactor compartment section
Ice load sensing
Improved simulated annealing algorithm
Reducing ill-posedness
Avoid local optima

ABSTRACT

The distributed ice load exerted on an engineering structure in a cold region can occasionally be treated as a static or dynamic problem. Examples of this are the thermal expansion load of the ice layer on a floating nuclear power plant (FNPP), and the distributed ice pressure exerted on the bow of an icebreaker during a continuous ice breaking operation. However, such ice loads are difficult to measure directly. Accordingly, indirect ice load monitoring is an effective way to obtain the ice load with the input of structural monitoring data. The mathematical models used for ice load sensing are usually ill-posed, from which the accuracy and stability of the sensing results usually suffer. Although D-optimal design method (DOD), C-optimal design method (COD) and block C-optimal design method (BCOD) have been used to reduce the ill-posedness of a mathematical model, there are certain disadvantages in employing DOD, COD and BCOD methods. These methods mentioned above are prone to falling into local optima, resulting in local far-field monitoring and poor optimization performance, especially when the mathematical model dimensions are large. An improved simulated annealing algorithm (ISAA) was developed in this study to avoid local optima and described based on a specific case study—the sensing of ice load exerted on the reactor compartment section of a FNPP. Comparative analysis of ice load sensing results reveals that when the mathematical model has a large dimensionality, ISAA can yield much lower ill-posed mathematical models compared to the DOD, COD, and BCOD methods. The mathematical model optimized by ISAA exhibits higher robustness, leading to accurate sensing results.

1. Introduction

As countries continuously adjust and optimize their energy structures to promote green and low-carbon development strategies, both traditional fossil fuels and emerging energies such as wind, wave, and solar power are increasingly failing to meet daily demands. Consequently, Floating Nuclear Power Plants (FNPPs) have garnered significant attention from governments worldwide as a clean, efficient, and flexible offshore nuclear technology. Compared to traditional power generation methods, offshore nuclear power offers advantages such as environmental friendliness, low cost, and minimal weather dependency, suggesting a promising future for its application.

It is essential to integrate mobile small nuclear power plant technology with shipbuilding and marine engineering technologies to implement offshore nuclear power. This involves constructing a mobile floating platform equipped with a nuclear reactor and power generation

system, known as a Floating Nuclear Power Plant (FNPP). Small modular reactor (SMR) is the most important component of FNPP, and the technical route of SMR can be roughly divided into single generator set with light water SMR, multi generator sets with light water SMR, fourth generation nuclear energy system SMR, movable/transportable SMR, and micro modular reactor (MMR). ACP100 is the first land-based SMR built in China, which is a type of small pressurized water reactor. The FNPP being designed in China will adopt technology similar to ACP100, and the FNPP will be deployed in the Bohai Sea near Penglai City, Shandong Province. The base is located approximately 11.46 km southwest of Penglai City, about 41.84 km from Longkou City, 64.46 km from Yantai City, 66.65 km from Zhaoyuan City, and 55.47 km from Qixia City (all distances are straight-line measurements). The FNPP will be anchored in the Bohai Sea near Yantai and will freeze into the ice during the winter, enduring the compressive forces of the ice. These ice-induced compressive forces may pose a threat to the hull structure of the

* Corresponding author. College of Shipbuilding Engineering, Harbin Engineering University, Harbin, 150001, China.

E-mail address: zhangmeng@hrbeu.edu.cn (M. Zhang).

<https://doi.org/10.1016/j.oceaneng.2024.119350>

Received 25 May 2024; Received in revised form 13 September 2024; Accepted 22 September 2024

Available online 26 September 2024

0029-8018/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

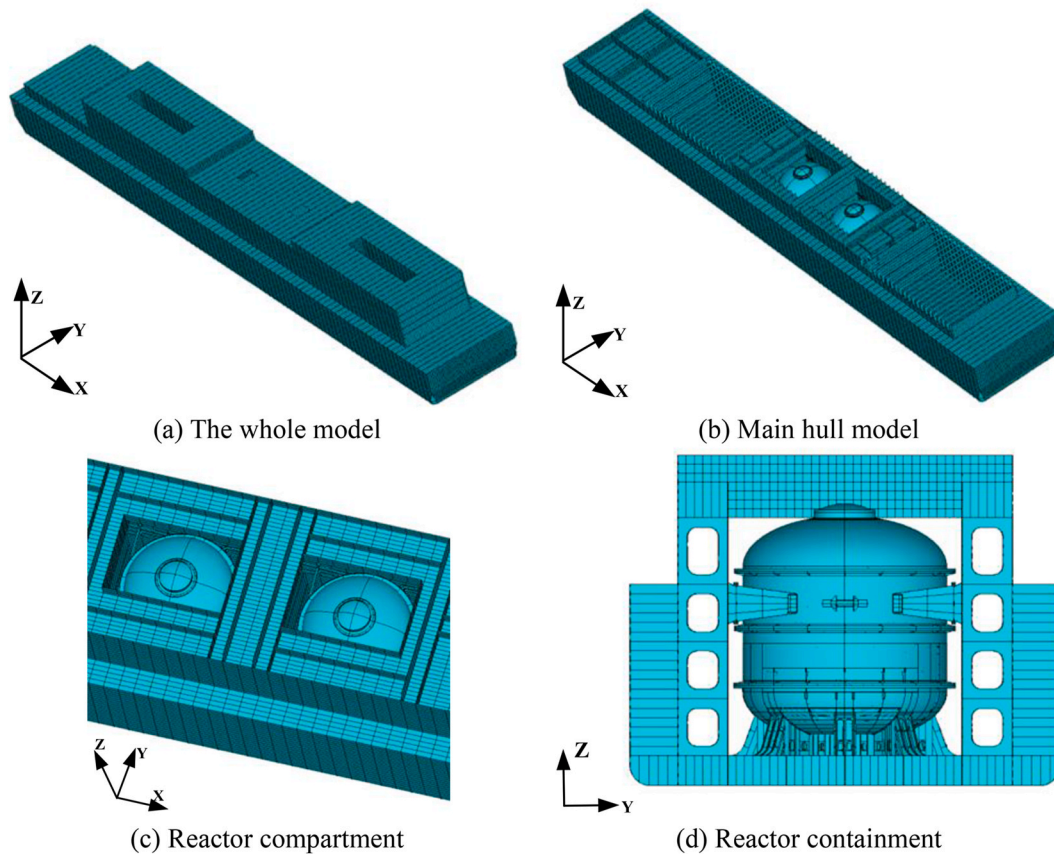


Fig. 1. The FNPP under research in China.

reactor compartment.

The impact of ice loads on engineering structures has been extensively studied through numerical simulations (Su et al., 2011; Wang et al., 2018; Zhou et al., 2016, 2019a, 2019b, 2020), model tests (Eik and Marchenko, 2010; Zhou et al., 2018), and ice load monitoring (Gagnon, 2008; Jeon et al., 2018; Lee et al., 2014; Ritch et al., 2008; Zhang et al., 2018, 2019a). However, these numerical simulation and model testing methods cannot accurately account for the various factors influencing ice loads, such as temperature, wind, water currents, ice thickness, snow cover, and ice type (Comfort et al., 2011; Gong et al., 1999; Kharik et al., 2015).

Ice load monitoring includes indirect monitoring (Jeon et al., 2018; Lee et al., 2014; Ritch et al., 2008; Zhang et al., 2018, 2019a) and direct monitoring (Gagnon, 2008). Indirect monitoring estimates ice loads by measuring the structural response and then performing a reverse calculation, whereas direct monitoring measures the ice loads applied to engineering structures using sensors. Direct ice load monitoring is typically used for fixed concrete structures, while Gagnon employed external impact panels to directly monitor ice loads on hull structures (Gagnon, 2008). In comparison, indirect monitoring methods are more suitable for long-term monitoring.

The bow of an icebreaking vessel possesses characteristics of high strength and rigidity, with locally structured inherent frequencies being notably high. Liu conducted an analysis on the inherent frequencies of the local structure of the Antarctic icebreaker S. A. Agulhas, revealing the lowest inherent frequency to be approximately 230 Hz (Liu, 2017). Furthermore, the maximum frequency of ice loads exerted on the hull structure is approximately 2.5 Hz (Liu, 2017; St John et al., 1990). Hence, ice loads can be regarded as a statically distributed ice pressure. Ritch et al. (2008) sensed the distribution of ice loads at localized positions on icebreaking vessels by measuring structural strains. Jeon et al. (2018) estimated localized ice pressures on the icebreaking vessel

ARAON using shear strain analysis methods, while Lee et al. (2014) also monitored structural strains on the ARAON, deriving the distribution of ice pressures.

Molikpaq is a nearshore ocean platform located in a cold region, which is subject to ice sheet compression and crushed ice impact during the freezing period, which has been extensively studied. Gagnon (2012) conducted detailed observations and analysis on the collision event between the ice layer and Molikpaq that occurred on May 12, 1986, with a focus on analyzing the relationship between ice spalling and structural vibration. Additionally, Molikpaq has a comprehensive monitoring system for ice load, structural strain, and structural deformation (Frederking and Sudom, 2006). The ice load can be directly measured using MEDOF panels, but this measurement method is suitable for ice loads with low frequency variations. In addition, ice load can also be indirectly monitored using the structural strain and deformation obtained from monitoring system. The core idea of indirect monitoring of ice load is to establish a relationship between structural strain or deformation and the total ice force, and then obtain the total ice force through simple calculations with the input of structural strain or deformation. Timco and Johnston analyzed the positive correlation between ice line force and the thickness of ice layer based on the monitoring results of ice load (Timco and Johnston, 2003).

One challenge in load sensing is the ill-posed nature of mathematical models. Conventional regularization methods such as Tikhonov regularization and truncated singular value decomposition can address ill-posed problems, but may be ineffective in certain cases. To achieve more stable and accurate sensing results, various improved regularization methods have been developed (Bazán and Boos, 2018; Calvetti et al., 2005; Hochstenbach and Reichel, 2012; Landi and Loli Piccolomini, 2010; Liu, 2012; Loubes and Pelletier, 2008; Morigi et al., 2006; Reichel and Sadok, 2008; Reichel and Shyshkov, 2010; Rezghi and Hosseini, 2009).

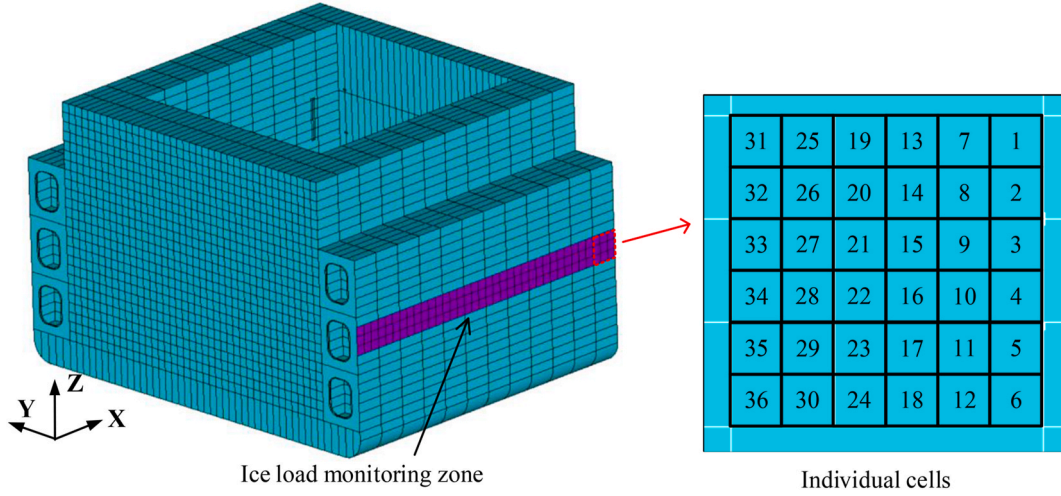


Fig. 2. Ice load monitoring zone of the rear reactor compartment section of a FNPP.

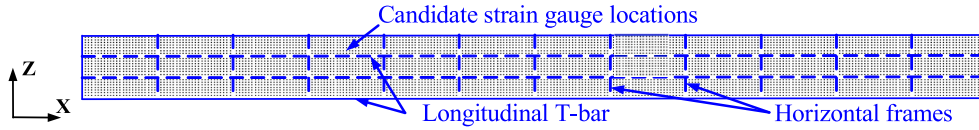


Fig. 3. Candidate strain gauge locations.

Some researchers have attempted to find effective and computationally cost-effective methods to overcome the ill-posed nature of load sensing mathematical models. Papadimitriou (2004) developed the Backward Sequential Sensor Placement (BSSP) algorithm for optimizing sensor locations to estimate structural dynamic parameters. Gupta and Dhingra (2013) utilized the D-Optimal Design (DOD) method to optimize strain gauge locations and reduce ill-posedness. Zhang et al. (2019b) proposed the C-Optimal Design (COD) method to reduce the ill-posedness of mathematical models by optimizing strain gauge locations.

However, the above-mentioned methods are prone to getting stuck in local optima due to issues with search strategies. Additionally, some algorithms exhibit extremely low computational efficiency when dealing with high-dimensional mathematical models, rendering them unusable. To enhance computational efficiency and optimize results, this study employs Simulated Annealing Algorithm (SAA) to optimize strain gauge locations and proposes an Improved Simulated Annealing Algorithm (ISAA). ISAA achieves superior optimization results and bolsters algorithmic stability by incorporating local search and multi-start search strategies. The effectiveness of the algorithm is validated through ice load sensing results obtained under three different operating conditions. Furthermore, a comparison of the overall relative error of ice load sensing results at different noise levels was conducted to analyse the robustness of ice load sensing mathematical models obtained through different optimization algorithms.

2. Existing works

DOD (Li et al., 2024), COD (Zhu et al., 2024) and BCOD (Zhang et al., 2019b) methods have been applied to reduce the ill-posedness of a mathematical model used for the sensing of a static distributed ice pressure exerted on engineering structures. In the applications of DOD, COD and BCOD methods, three main steps are applied: establishing a mathematical model, optimizing the strain gauge locations, and outputting the optimization results. The first step (establishing a mathematical model) is the same between DOD, COD and BCOD methods, as is the last step (outputting the optimization results).

2.1. Establishing a mathematical model

The length of FNPP is 230m, the width is 36m, and the draft is 10m. The displacement of the icebreaker is approximately 83,000 tons. The FNPP was constructed in ANSYS using shell elements (shell181) and beam elements (beam188). The material parameters were defined according to the properties of steel. The finite element model (FEM) of the FNPP is depicted in Fig. 1.

The rear reactor compartment section of a FNPP was selected as a case study, as shown in Fig. 2.

The ice load monitoring zone was divided into several individual cells, and it was assumed that the ice load within each cell is uniform. The area of the ice load monitoring zone is approximately 69.12 m², and was almost evenly divided into 432 individual cells, as shown in Fig. 2. The area of each individual cell is approximately 0.16 m².

The nodes of the structural finite element model (FEM) within the monitoring zone were used as candidate strain gauge locations, as shown in Fig. 3. In addition, the nodes of the beams and frames are not included in the candidate strain gauge locations, nor are the nodes of the hull plate close to beams and frames.

Finally, a mathematical model, representing the linear mapping relationship between the approximated uniform ice load within each cell and the structural response at every candidate strain gauge location, was numerically established. The specific procedures were described in detail by Zhang et al. (2019b). The mathematical model can be expressed as a linear system of equations, as shown in Eq. (1).

$$\begin{Bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{Bmatrix} = \begin{bmatrix} K_{11} & K_{12} & \cdots & K_{1m} \\ K_{21} & K_{22} & \cdots & K_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ K_{n1} & K_{n2} & \cdots & K_{nm} \end{bmatrix} \cdot \begin{Bmatrix} f_1 \\ f_2 \\ \vdots \\ f_m \end{Bmatrix} \quad (1)$$

Eq. (1) can be simplified as:

$$Kf = e \quad (2)$$

where.

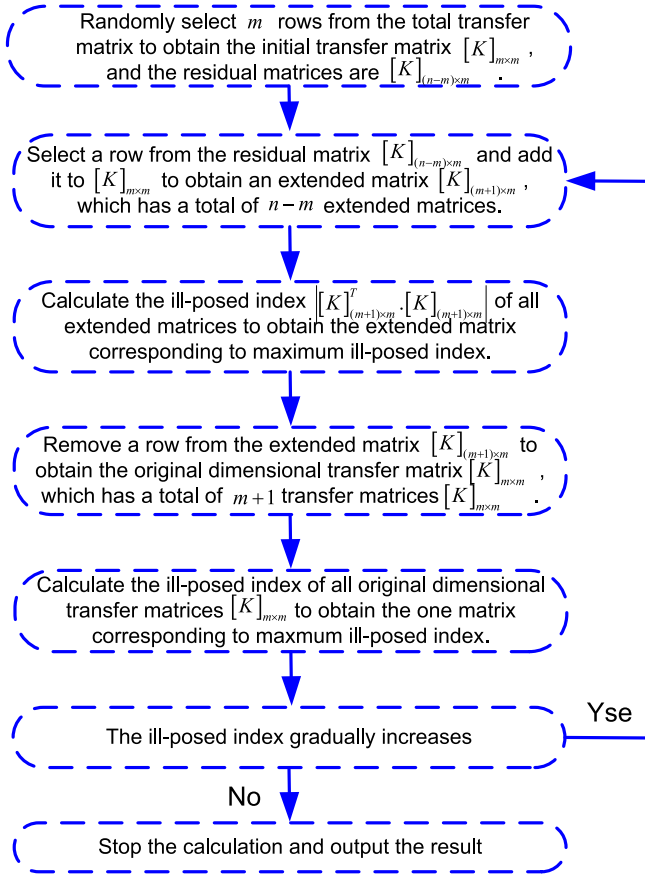


Fig. 4. Flowchart of DOD method.

- (1) $\mathbf{K} = (K_{ij})_{n \times m}$ is the total transfer matrix;
- (2) $\mathbf{f} = (f_i)_{m \times 1}$ is the load vector;
- (3) $\mathbf{e} = (e_{ij})_{n \times 1}$ is the strain vector;
- (4) e_i is the structural micro-strain at candidate location No. i ;
- (5) n is the number of candidate locations;
- (6) f_i is the uniform ice pressure of cell No. i ;
- (7) m is the number of cells.

There are 10692 candidate strain gauge locations, as shown in Fig. 3. Therefore, the total transfer matrix (obtained numerically) can be expressed as $\mathbf{K} = (K_{ij})_{10692 \times 432}$. It is clearly impossible to arrange the strain gauges at all candidate locations. Therefore, it is reasonable to select several candidate locations, p , where the strain gauges are arranged. The total transfer matrix $\mathbf{K} = (K_{ij})_{n \times m}$ is then reduced to $\mathbf{K} = (K_{ij})_{p \times m}$.

2.2. D-optimal design method

When the ice load monitoring zone is divided into m individual cells and there are n candidate strain gauge locations, the total transfer matrix can be expressed as $\mathbf{K} = (K_{ij})_{n \times m}$. Aiming at determining m optimal strain gauge locations to arrange the strain gauges, i.e. obtaining the transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$, the DOD method can be used. In the application of this method, the determinant of $\mathbf{K}^T \cdot \mathbf{K}$ ($\mathbf{K} = (K_{ij})_{m \times m}$), i.e. $|\mathbf{K}^T \cdot \mathbf{K}|$, is used as the objective function, and a large $|\mathbf{K}^T \cdot \mathbf{K}|$ value indicates that the mathematical model has low ill-posedness. The objective of the DOD method is to maximise $|\mathbf{K}^T \cdot \mathbf{K}|$.

The main procedures of this method are as follows.

Step 1:

Select m rows from the total transfer matrix $\mathbf{K} = (K_{ij})_{n \times m}$ at random to generate the initial transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$. Then, removing the randomly selected m rows from the total transfer matrix $\mathbf{K} = (K_{ij})_{n \times m}$ will result in the residual transfer matrix $\mathbf{K} = (K_{ij})_{(n-m) \times m}$.

In this step, the total transfer matrix $\mathbf{K} = (K_{ij})_{n \times m}$ is divided into 2 transfer matrix, i.e. the initial transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$ and the residual transfer matrix $\mathbf{K} = (K_{ij})_{(n-m) \times m}$.

Step 2:

Add one row of the residual transfer matrix $\mathbf{K} = (K_{ij})_{(n-m) \times m}$ to the initial transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$ one by one and the extended transfer matrices $\mathbf{K} = (K_{ij})_{(m+1) \times m}$ can be obtained. There are $n-m$ extended transfer matrices $\mathbf{K} = (K_{ij})_{(m+1) \times m}$. Then, calculate the ill-posed index $|\mathbf{K}^T \cdot \mathbf{K}|$ ($\mathbf{K} = (K_{ij})_{(m+1) \times m}$) of each extended transfer matrix $\mathbf{K} = (K_{ij})_{(m+1) \times m}$. Thereafter, the extended transfer matrix $\mathbf{K} = (K_{ij})_{(m+1) \times m}$ corresponding to the maximum ill-posed index $|\mathbf{K}^T \cdot \mathbf{K}|$ is selected and treated as the result of step 2.

For example, the first row of the residual transfer matrix $\mathbf{K} = (K_{ij})_{(n-m) \times m}$ is added to the initial transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$. Then, the first extended transfer matrix $\mathbf{K} = (K_{ij})_{(m+1) \times m}$ can be obtained. Similarly, the second row of the residual transfer matrix $\mathbf{K} = (K_{ij})_{(n-m) \times m}$ is added to the initial transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$. Then, the second extended transfer matrix $\mathbf{K} = (K_{ij})_{(m+1) \times m}$ can be obtained. With this approach, $n-m$ extended transfer matrices $\mathbf{K} = (K_{ij})_{(m+1) \times m}$ can be obtained. Thereafter, $n-m$ determinants of $\mathbf{K}^T \cdot \mathbf{K}$ ($\mathbf{K} = (K_{ij})_{(m+1) \times m}$) can be calculated. Finally, the extended transfer matrix $\mathbf{K} = (K_{ij})_{(m+1) \times m}$ corresponding to the largest determinant of $\mathbf{K}^T \cdot \mathbf{K}$ ($\mathbf{K} = (K_{ij})_{(m+1) \times m}$) can be gained.

Step 3:

Exclude one row from the extended transfer matrix $\mathbf{K} = (K_{ij})_{(m+1) \times m}$ obtained in step 2 one by one and the original dimensional transfer matrices $\mathbf{K} = (K_{ij})_{m \times m}$ can be obtained. There are $m+1$ original dimensional transfer matrices $\mathbf{K} = (K_{ij})_{m \times m}$. Then, calculate the ill-posed index $|\mathbf{K}^T \cdot \mathbf{K}|$ ($\mathbf{K} = (K_{ij})_{m \times m}$) of each original dimensional transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$. Thereafter, the original dimensional transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$ corresponding to the maximum ill-posed index $|\mathbf{K}^T \cdot \mathbf{K}|$ is selected and treated as the result of step 3.

For example, delete the first row of the extended transfer matrix $\mathbf{K} = (K_{ij})_{(m+1) \times m}$ and the first original dimensional transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$ can be obtained. Similarly, delete the second row of the extended transfer matrix $\mathbf{K} = (K_{ij})_{(m+1) \times m}$ and the second original dimensional transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$ can be obtained. With this approach, $m+1$ original dimensional transfer matrices $\mathbf{K} = (K_{ij})_{m \times m}$ can be obtained. Thereafter, $m+1$ determinants of $\mathbf{K}^T \cdot \mathbf{K}$ ($\mathbf{K} = (K_{ij})_{m \times m}$) can be calculated. Finally, the original dimensional transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$ corresponding to the largest determinant of $\mathbf{K}^T \cdot \mathbf{K}$ ($\mathbf{K} = (K_{ij})_{m \times m}$) can be gained. At this step, the transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$ is updated once.

Step 4:

Repeat steps 2 and 3 until the determinant $\mathbf{K}^T \cdot \mathbf{K}$ ($\mathbf{K} = (K_{ij})_{m \times m}$) no

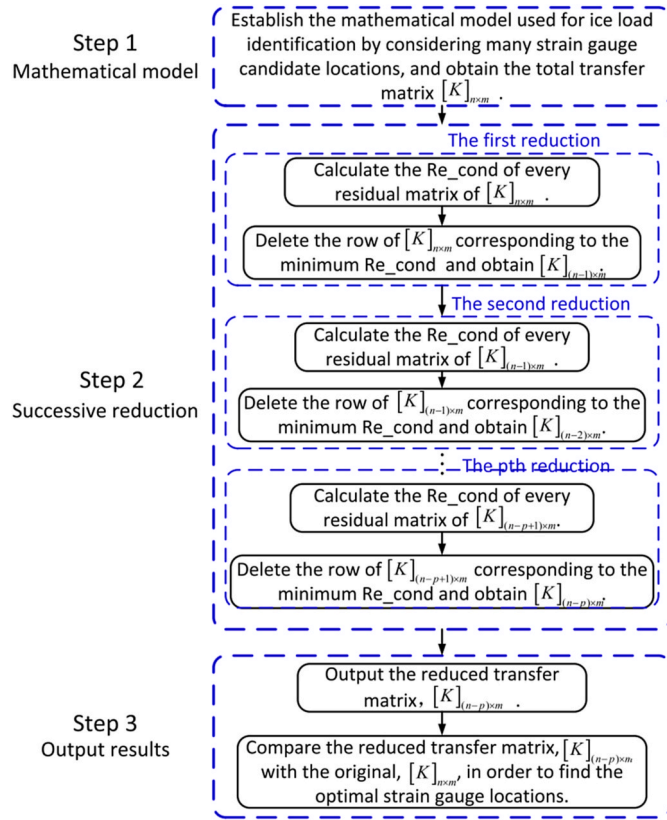


Fig. 5. Flowchart of COD method (Zhang et al., 2019b).

longer increases, and the final transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$ can be obtained.

Step 5:

Compare the final transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$ with the total transfer matrix $\mathbf{K} = (K_{ij})_{n \times m}$ to determine the optimal strain gauge locations.

The DOD method can be illustrated by the flowchart shown in Fig. 4.

2.3. C-optimal design method

In the application of the COD method, the condition number of the transfer matrix is used as the objective function and an ill-posed index, and a small condition number means the mathematical model has low ill-posedness. The objective of the COD method is to minimise the condition number of the transfer matrix.

The two-norm condition number of a matrix $\mathbf{K} = (K_{ij})_{m \times n}$ can be given by Eq. (3) when $m = n$.

$$\text{cond}(\mathbf{K}) = \|\mathbf{K}\| \cdot \|\mathbf{K}^{-1}\| \quad (3)$$

where $\|\mathbf{K}\|$ is the two-norm of $\mathbf{K} = (K_{ij})_{m \times n}$. If $m \neq n$, the two-norm condition number can be expressed as:

$$\text{cond}(\mathbf{K}) = \|\mathbf{K}\| \cdot \|\mathbf{K}^+\| \quad (4)$$

where $\mathbf{K}^+$ is the Moore–Penrose inverse matrix of $\mathbf{K} = (K_{ij})_{m \times n}$.

The main procedures of the COD method are described as follows.

Step 1:

Exclude one row from the total transfer matrix $\mathbf{K} = (K_{ij})_{n \times m}$ will result in the residual transfer matrix $\mathbf{K} = (K_{ij})_{(n-1) \times m}$. There are n re-

sidual transfer matrices $\mathbf{K} = (K_{ij})_{(n-1) \times m}$. Then, calculate the condition number of each residual transfer matrix $\mathbf{K} = (K_{ij})_{(n-1) \times m}$. Thereafter, the residual transfer matrix $\mathbf{K} = (K_{ij})_{(n-1) \times m}$ corresponding to the minimum condition number is selected and treated as the result of step 1.

For example, if only the first row of the total transfer matrix $\mathbf{K} = (K_{ij})_{n \times m}$ is excluded, then the transfer matrix $\mathbf{K} = (K_{ij})_{n \times m}$ is transformed into $\mathbf{K} = (K_{ij})_{(n-1) \times m}$, which is the first residual transfer matrix. If only the second row of the total transfer matrix $\mathbf{K} = (K_{ij})_{n \times m}$ is excluded, then the transfer matrix $\mathbf{K} = (K_{ij})_{n \times m}$ is transformed into $\mathbf{K} = (K_{ij})_{(n-1) \times m}$, which is the second residual transfer matrix. With this approach, n residual transfer matrices $\mathbf{K} = (K_{ij})_{(n-1) \times m}$ can be obtained. Thereafter, n condition numbers of $\mathbf{K} = (K_{ij})_{(n-1) \times m}$ can be calculated. Finally, the residual transfer matrix $\mathbf{K} = (K_{ij})_{(n-1) \times m}$ corresponding to the minimum condition number of $\mathbf{K} = (K_{ij})_{(n-1) \times m}$ can be gained. The transfer matrix $\mathbf{K} = (K_{ij})_{n \times m}$ becomes $\mathbf{K} = (K_{ij})_{(n-1) \times m}$ in this step, which is the first reduction operation.

Step 2:

The residual transfer matrix $\mathbf{K} = (K_{ij})_{(n-1) \times m}$ obtained in step 1 after the first reduction operation is used as the total transfer matrix of the step 2. If only the first row of the total transfer matrix $\mathbf{K} = (K_{ij})_{(n-1) \times m}$ is excluded, then the transfer matrix $\mathbf{K} = (K_{ij})_{(n-1) \times m}$ is transformed into $\mathbf{K} = (K_{ij})_{(n-2) \times m}$, which is the first residual transfer matrix. If only the second row of the total transfer matrix $\mathbf{K} = (K_{ij})_{(n-1) \times m}$ is excluded, then the transfer matrix $\mathbf{K} = (K_{ij})_{(n-1) \times m}$ is transformed into $\mathbf{K} = (K_{ij})_{(n-2) \times m}$, which is the second residual transfer matrix. With this approach, $n - 1$ residual transfer matrices $\mathbf{K} = (K_{ij})_{(n-2) \times m}$ can be obtained. Thereafter, $n - 1$ condition numbers of $\mathbf{K} = (K_{ij})_{(n-2) \times m}$ can be calculated. Finally, the residual transfer matrix $\mathbf{K} = (K_{ij})_{(n-2) \times m}$ corresponding to the minimum condition number of $\mathbf{K} = (K_{ij})_{(n-2) \times m}$ can be gained. The transfer matrix $\mathbf{K} = (K_{ij})_{(n-1) \times m}$ becomes $\mathbf{K} = (K_{ij})_{(n-2) \times m}$ in this step, which is the second reduction operation.

Step 3:

The reduction operation is repeated until the number of matrix rows decreases to m .

Step 4:

The final transfer matrix $\mathbf{K} = (K_{ij})_{m \times m}$ is compared with the original transfer matrix $\mathbf{K} = (K_{ij})_{n \times m}$ to determine the optimal strain gauge locations.

The COD method deals with calculating and improving the condition number of the transfer matrix in a direct manner. Therefore, the computational speed of the method is slow compared with the DOD method.

The COD method can be illustrated by the flowchart shown in Fig. 5.

2.4. Block C-optimal design method

To alleviate the computational cost of the COD method, the BCOD method was developed. In the application of the BCOD method, the total transfer matrix is initially divided into several sub-matrices. Each sub-matrix is then reduced to a suitable size using the COD method. Finally, all reduced sub-matrices are combined into one transfer matrix, and the combined matrix is then further reduced to $\mathbf{K} = (K_{ij})_{m \times m}$, using COD method.

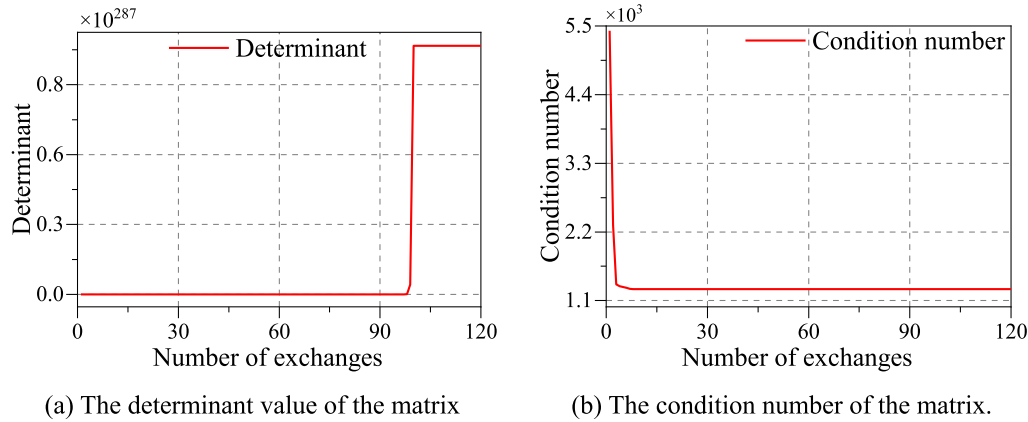


Fig. 6. The variation of the matrix ill-posedness index during the optimization process using the DOD method.

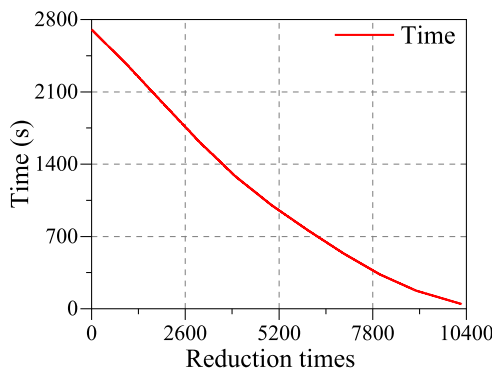


Fig. 7. Computational time of the COD method.

2.5. Optimization results using existing works

The strain gauge installation positions are optimized using the aforementioned methods separately. The optimization process using the DOD method is illustrated in Fig. 6.

The results in Fig. 6 indicate that the DOD method exhibits high computational efficiency, but the optimization effectiveness is limited. The resulting condition number of the matrix is 1285, indicating a high degree of ill-posedness, making it impossible to establish a stable mathematical model for ice load sensing.

The number of candidate strain gauge positions is 10,692, with a target strain gauge count of 432. The COD method requires 10260 reduction steps to obtain the optimization results. The first reduction step involves computing and comparing the condition number of the matrix 10,692 times. Subsequent reduction steps require one less comparison each time, i.e., 10,691 times for the second reduction, and so forth. Due to the varying computational complexity and matrix dimensions in each reduction step, the computation time differs for each step. We simulate and calculate the time consumption for the 1st, 1000th, 2000th, 3000th, 4000th, 5000th, 6000th, 7000th, 8000th, 9000th, and 10260th reduction steps, and employ linear interpolation to estimate the computation time for the remaining steps, as illustrated in Fig. 7.

The total time required for the COD method, summing the computation time of each reduction step in Fig. 5, is 131.7 days. It is impractical to apply the COD method to optimization involving candidate strain gauge positions exceeding 10,000, given the excessive time consumption. Therefore, the COD method is not suitable for optimization in this scenario.

When employing BCOD for measurement point optimization, the

total candidate points are divided into five blocks, resulting in five sub-matrices. Each sub-matrix has 2140 rows, except for the last one, which has 2132 rows. The optimization process for each sub-matrix and the combined matrix is illustrated in Fig. 8.

The BCOD method adopts a strategy of parallel computation with block matrices, leading to a significant improvement in computational efficiency compared to the COD method. The results in Fig. 8 demonstrate that the BCOD method reduces the matrix condition number to 259 after optimization, indicating a noticeable improvement in ill-posedness compared to before optimization. However, it's not entirely eliminated, thus still unable to guarantee the stability of ice load sensing. When the number of candidate strain gauge positions exceeds 10,000, existing optimization methods face limitations in computational efficiency and search strategies, making practical application very time-consuming or susceptible to local optimal solutions. In scenarios with a very large number of measurement points, a method that can simultaneously address computational efficiency and optimization effectiveness is needed.

3. Improved simulated annealing algorithms

Simulated Annealing Algorithm (SAA) is a heuristic optimization technique employed to quickly search for the optimal value of complex analytical functions, thereby optimizing the objective function. The SAA originates from the principle of solid annealing, which is a probability-based algorithm that heats a solid to a sufficient height and then slowly cools it. When heated, the particles inside the solid become disordered with the temperature rise, and the internal energy increases. When slowly cooled, the particles gradually become ordered, reaching equilibrium at each temperature, and finally reaching the ground state at room temperature, where the internal energy decreases to a minimum. The SAA leverages this analogy from the physical process to address optimization problems. It is a generalised probabilistic method aimed at searching for solutions close to the optimum within a wide search space.

3.1. Simulated annealing algorithms

3.1.1. Algorithm steps

The SAA simulates the cooling process from a higher initial temperature to a lower temperature, enhancing global search capabilities and solution space exploration by accepting some suboptimal solutions due to high-temperature instability. The specific steps are as follows.

Step 1: Parameter Initialization

- Randomly select an initial solution (randomly generate a set of strain gauge position combinations);

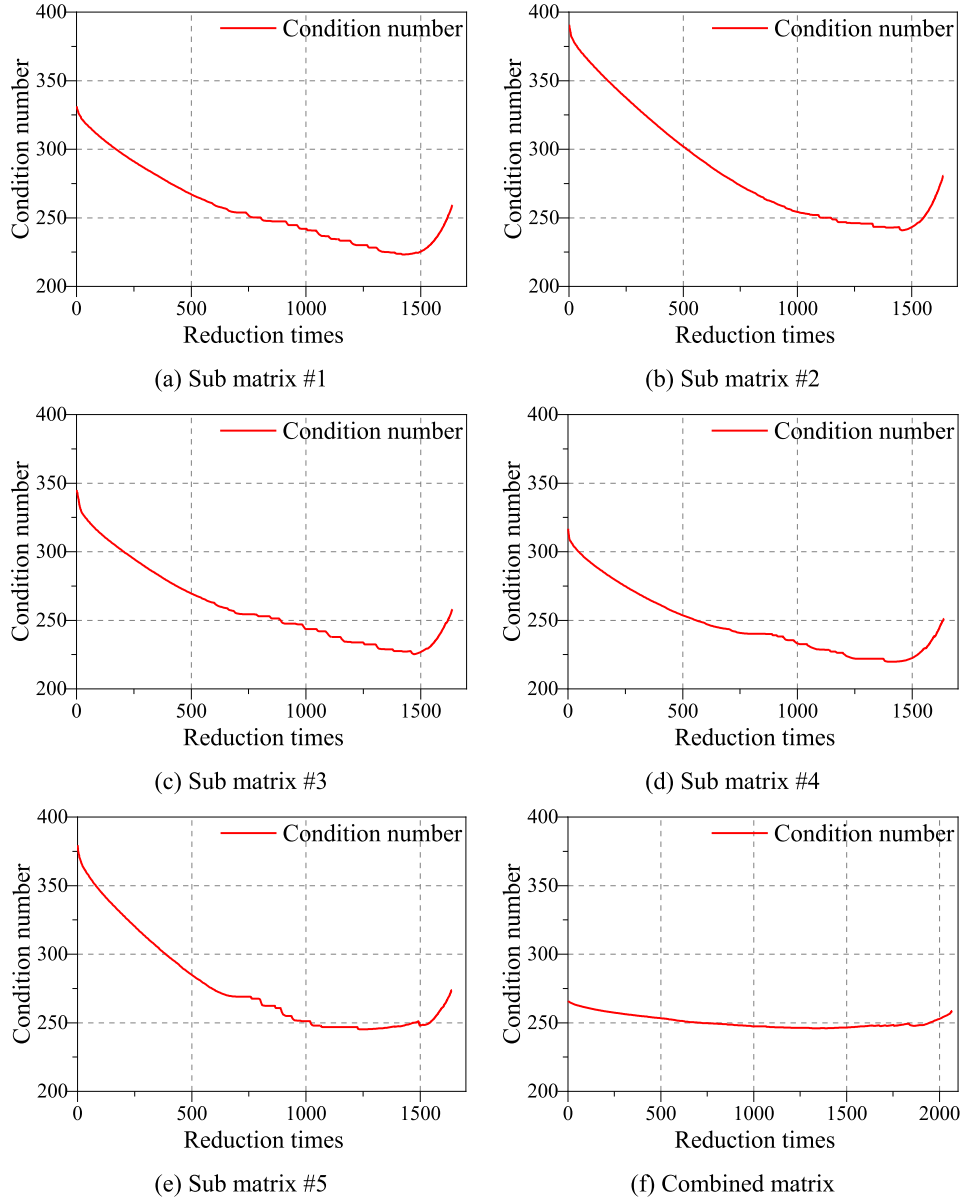


Fig. 8. The variation of the matrix ill-posedness index during the optimization process using the BCOD method.

- b. Set the condition number of the transfer matrix as the objective function;
- c. Set initial temperature, cooling rate, and termination conditions (maximum number of iterations or minimum temperature).

Step 2: Iterative calculation

- a. Generate a new solution randomly;
- b. Calculate the change in the objective function value between the new solution and the current solution;
- c. Determine whether to accept the new solution based on the Metropolis criterion;
- d. Gradually decrease the system's temperature according to the cooling rate.

Step 3: Termination

When the temperature reaches the minimum temperature or the maximum number of iterations is reached, exit the loop and obtain the global optimal solution.

The SAA can be illustrated by the flowchart shown in Fig. 9.

3.1.2. Metropolis criterion

The Metropolis criterion originates from the Metropolis algorithm in statistical mechanics, initially applied to randomly selecting samples in Monte Carlo methods. In optimization problems, the Metropolis criterion allows for the acceptance of suboptimal solutions, aiding in breaking free from local optima and increasing the likelihood of discovering the global optimum. The decision-making process is outlined as follows.

(1) Calculation of Objective Function Difference:

Calculate the difference ΔE in the objective function between the new solution s' and the current solution s :

$$\Delta E = f(s') - f(s) \quad (5)$$

$$f(s) = \text{cond}(K_s) \quad (6)$$

where K_s represents the transfer matrix corresponding to solution s .

For minimization problems, if $\Delta E < 0$ (i.e., the new solution is bet-

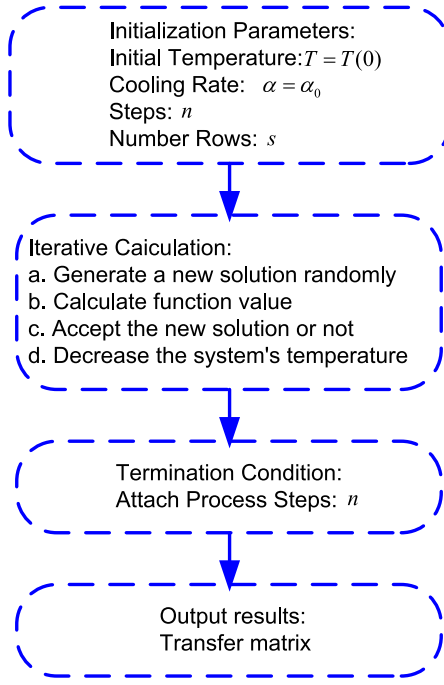


Fig. 9. Flowchart of SAA

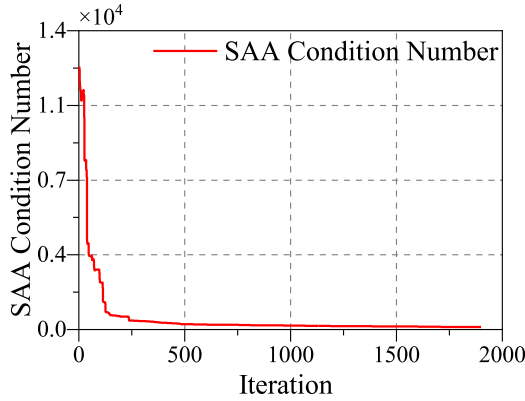


Fig. 10. Changes in matrix condition number during the optimization process using SAA.

ter), it is immediately accepted.

(2) Calculation of Acceptance Probability:

If $\Delta E > 0$ (i.e., the new solution is worse), compute the acceptance probability P using the Metropolis criterion:

$$P = e^{-\frac{\Delta E}{T}} \quad (7)$$

where T denotes the current temperature iteration.

(3) Random Decision:

Generate a random number r between 0 and 1. If $r \leq P$, accept the new solution; otherwise, keep the current solution.

The characteristic of the Metropolis criterion is that at higher temperatures, suboptimal solutions are occasionally accepted, facilitating a more extensive exploration of the solution space and avoiding premature convergence to local optima.

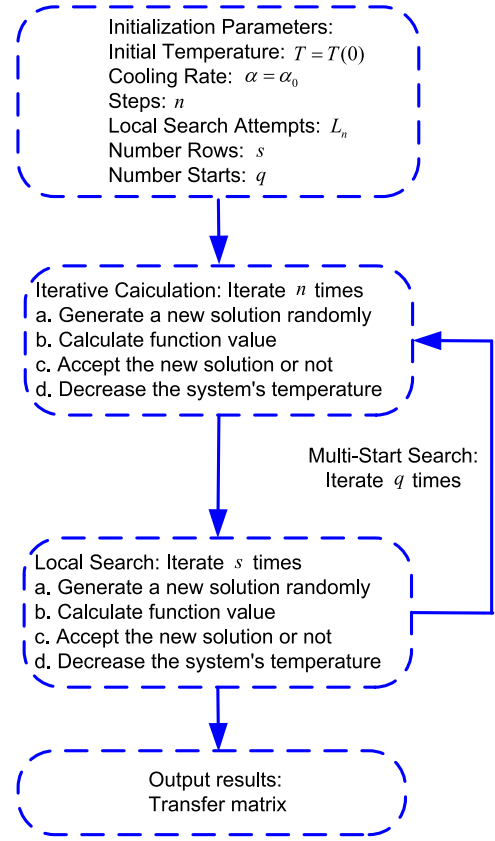


Fig. 11. Flowchart of ISAA

3.1.3. Optimization results

In optimizing the strain gauge positions using SAA, with a target number of 432 strain gauges, the initial temperature is set to 10000, the cooling rate is 0.93, and the termination condition is set to iterate 1900 times. The changes in the matrix condition number during the optimization process using SAA are depicted in Fig. 10.

The results in Fig. 10 indicate that the matrix condition number reaches 112.5 after optimization, with the entire optimization process taking 260 s. Although SAA has fewer parameters and the reasonable parameter range is easy to determine, precise parameters require multiple computations to determine accurately. The initial solution significantly influences the optimization results, and in cases where the number of candidate strain gauge positions is large, there may still be a risk of converging to a local optimum.

3.2. Improved simulated annealing algorithms

As a stochastic optimization algorithm, the optimization outcomes of SAA are inherently unstable. Therefore, an improved Simulated Annealing Algorithm (ISAA) is proposed to enhance the stability of the algorithm and reduce its sensitivity to the initial solution. This improvement introduces local search algorithms and multi-start search strategies based on the SAA. These enhancements aim to expand the exploration range of the solution space and increase the stability of the algorithm's results. The specific implementation steps are as follows.

(1) Parameter Initialization

- Set the initial temperature T ;
- Define the termination conditions for simulated annealing (minimum temperature or maximum number of iterations);
- Set the cooling rate α ;
- Define the local search termination parameter L_n ;

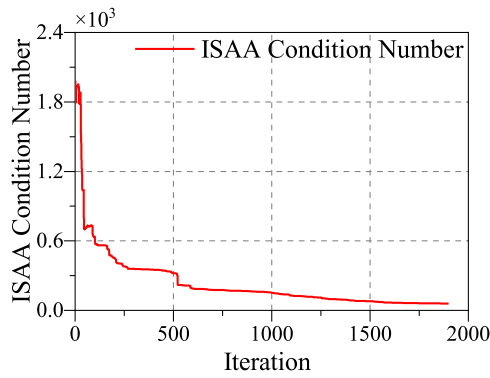


Fig. 12. Changes in matrix condition number during the optimization process using ISAA.

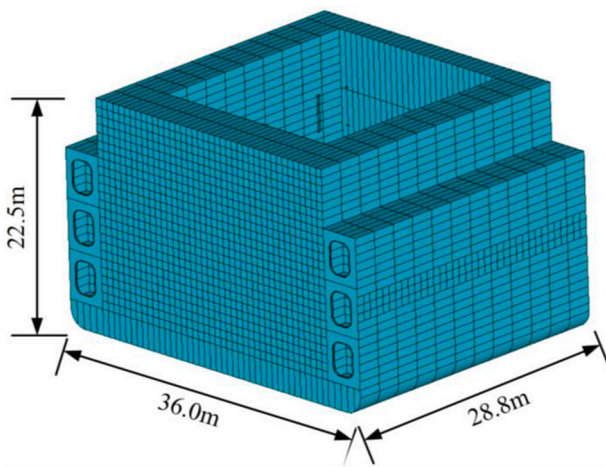


Fig. 13. FEM of the reactor containment segment of the FNPP.

e. Set the number of starting points q .

(2) Multi-Start Search

Generate multiple initial solutions based on the number of starting

points. For each starting point, randomly generate an initial solution s , i. e., randomly select s measurement point positions to obtain the corresponding transfer matrix. The objective function is the condition number of the transfer matrix.

(3) Simulated Annealing Iterative Process:

- Randomly generate a new solution s' .
- Calculate the objective function value of the new solution.
- Accept or reject the new solution based on the Metropolis criterion.
- Decrease the temperature T according to the cooling rate α , $T = \alpha * T$.
- Repeat the above steps until the termination conditions for simulated annealing are met, resulting in the current solution.

(4) Local Search Iterative Process

- Replace any measurement point position in the current solution with another position to obtain a neighboring solution s_l .
- Calculate the objective function value of s_l .
- If the objective function value of the neighboring solution is lower than that of the current solution, accept the neighboring solution and set the local search count L to 1. Otherwise, repeat the calculation for a new neighboring solution and increase the local search count $L = L + 1$, until $L = L_n$.

(5) Overall Iteration

Repeat steps (2) to (4) for all starting points to explore the solution space comprehensively and obtain the optimal solution.

The ISAA can be illustrated by the flowchart shown in Fig. 11.

By incorporating multi-start and local search strategies, the ISAA aims to improve the overall performance and reliability of the

Table 1

Main material parameters of the ice.

Solid element		Cohesive element	
Parameter	Value	Parameter	Value
Young's modulus (GPa)	5.00	Density (Kg/m^3)	895
Density (Kg/m^3)	910	Breaking energy (J/m^2)	5200
Poisson's ratio	0.37	Tensile strength (MPa)	1.04
Yield stress (MPa)	30.00	Shear strength (MPa)	1.04
Failure strain	0.7	Traction-separation law	Bilinear type

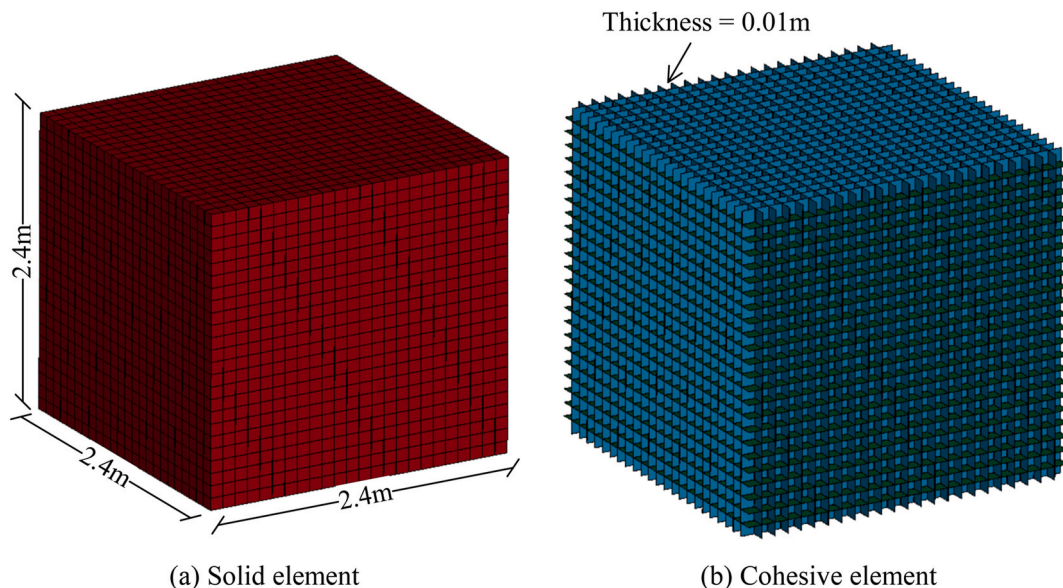


Fig. 14. Ice model.

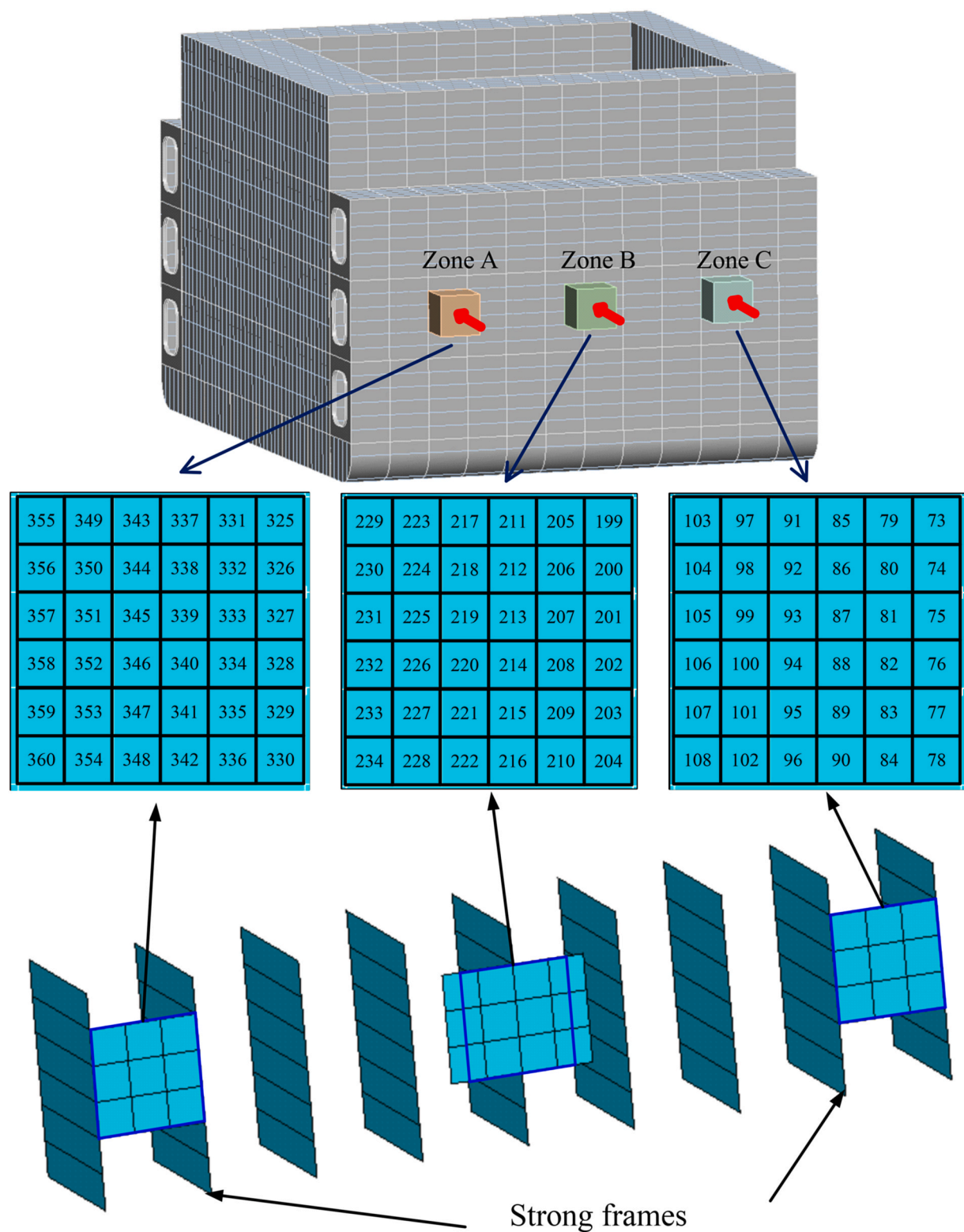


Fig. 15. Ice load affected zone.

Table 2
Ice load conditions.

Load case	Ice block quantity	Affected zones
1	1	Zone B
2	2	Zone A & Zone C
3	3	Zone A & Zone B & Zone C

optimization process, leading to more consistent and stable results.

The initial temperature is set to 10000, the cooling rate is 0.93, the target number of strain gauge positions is 432, the number of starting points is 5, and the consecutive local search attempts are set to 1000. The maximum number of iterations is set to 1900. Fig. 12 shows the changes in the matrix condition number during the optimization process using ISAA.

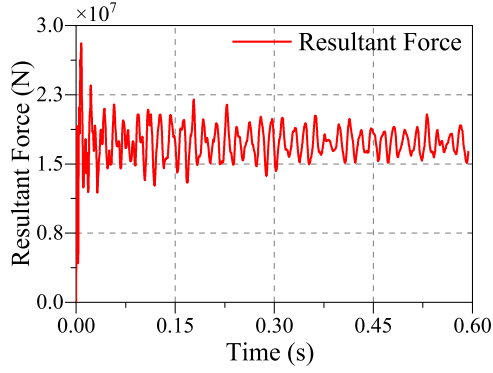


Fig. 16. Resultant force on the outer plate.

The results in Fig. 12 indicate that the matrix condition number reaches 57.1 after optimization, with the entire process taking 494 s. After the SAA process, ISAA performs a local search to find higher-quality solutions within the neighbourhood and starts the search from multiple initial solutions. This expands the exploration of the solution space, reduces dependency on a single initial solution, and increases the likelihood of discovering the global optimum. Although ISAA takes longer than SAA, it is more stable and yields better optimization results.

The mathematical model for ice load sensing established after optimizing the strain gauge installation positions is as follows:

$$(K)_{432 \times 432} \cdot (f)_{432 \times 1} = (\varepsilon)_{432 \times 1} \quad (8)$$

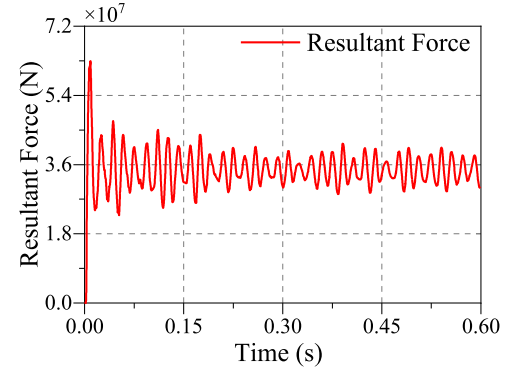


Fig. 19. Resultant force on the outer plate.

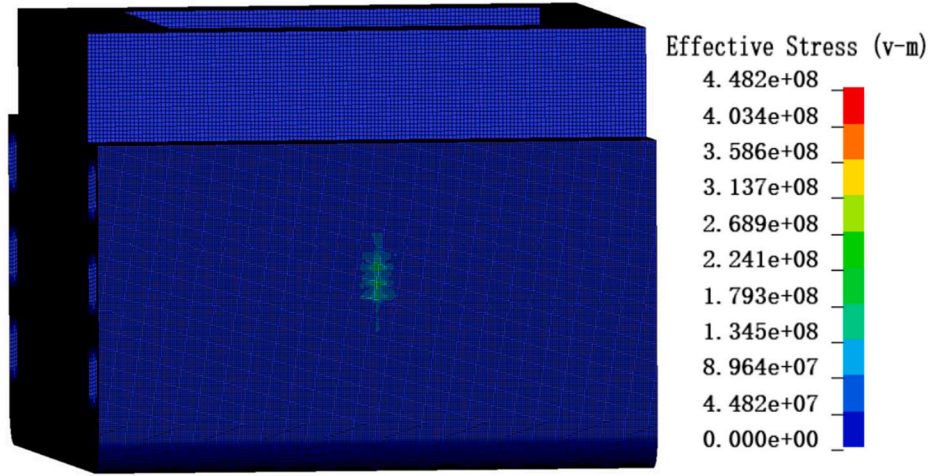


Fig. 17. Stress distribution in the containment segment.

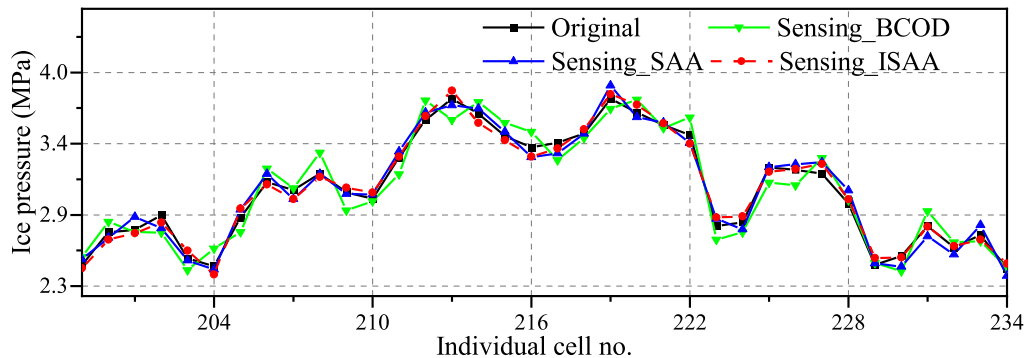


Fig. 18. Ice load sensing results (Zone B, Load case 1).

4. Numerical examples of ice load sensing

The study focuses on the reactor containment segment of the FNPP, simulating the ice pressure exerted on the segment's outer plate during ice compression. The optimized ice load sensing mathematical model is applied to sense the ice loads and validate the accuracy of the sensing model. The FEM of the reactor containment segment of the FNPP is illustrated in Fig. 13.

The outer plate of the reactor containment segment is made of high-strength steel with a thickness of 36 mm. The overall FEM is established

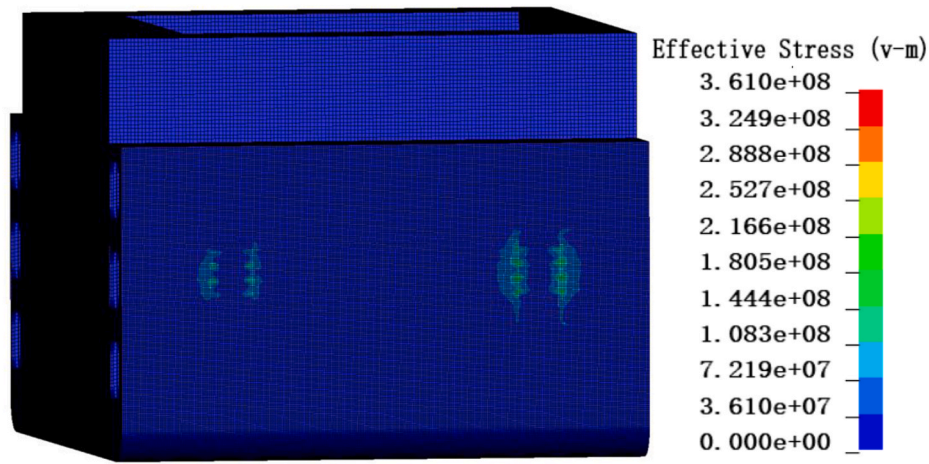
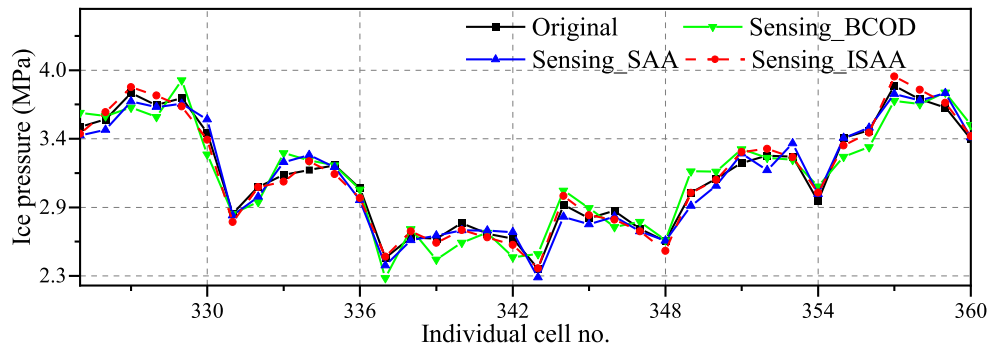
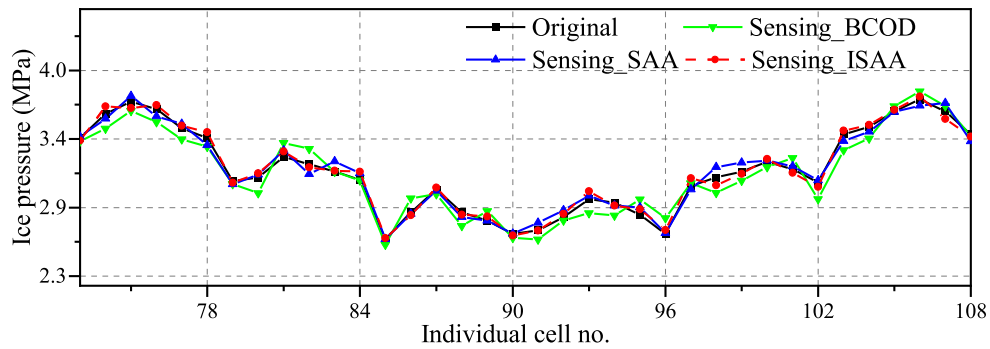


Fig. 20. Stress distribution in the containment segment.



(a) Zone A



(b) Zone C

Fig. 21. Ice load sensing results (Load case 2).

using shell181 elements with a size of 0.2m. In the ice compression area, the element size is reduced to 0.1m. The ice model is established using the cohesive element method, which requires a combination of solid elements and cohesive elements to form the material model. The maximum ice thickness in the Bohai Sea area of China is approximately 1.2 m. The ice thickness is doubled in the simulation of ship-ice interaction considering ice accumulation. Therefore, the ice thickness of 2.4 m is not a strict thickness. In addition, a huge ice layer was not set up to compress the hull, due to the high simulation cost of ship-ice interaction in LS-DYNA software, but instead set up three small ice blocks to compress the hull to verify the accuracy of the ice load sensing method developed in this study. The ice block size is set to 2.4m × 2.4m × 2.4m,

with solid element sizes of 0.1m and cohesive element thickness of 0.01m, as shown in Fig. 14.

The main material parameters of the ice are listed in Table 1.

The interaction between the ice and the ship's outer plate is simulated using LS-DYNA software. The surfaces on both sides of the containment vessel segment are fixed. The contact type between the ice block and the chamber is solid contact. The static friction coefficient is set to 0.2, and the dynamic friction coefficient is set to 0.1. There is an initial distance of approximately 0.01 m between the ice block and the outer plate of the ship. A constant force is applied behind the ice block as the compression force, and the overall calculation time is 0.6 s.

This FNPP is highly likely to be placed in the Bohai Sea area of China

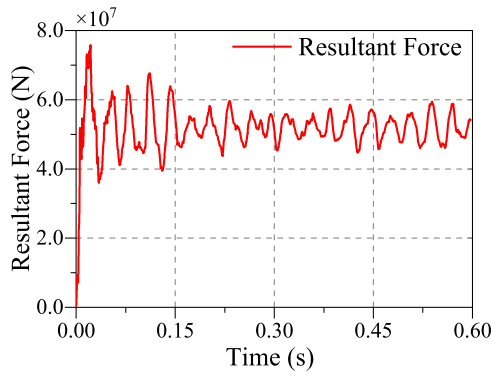


Fig. 22. Resultant force on the outer plate.

in the future. During the winter icing period, this ship will be surrounded by intact ice and subjected to static pressure from the ice. As the seasons change and temperatures rise, this ship will also be subjected to the impact of broken ice. When the ship is surrounded by intact ice layers and subjected to static compression from the ice layers, the ice load sensing can be treated as static mechanical problem, which is also the focus of this study. When this ship is impacted by broken ice, the ice load sensing may be a static or dynamic mechanical problem. In this study, a pressure of 3 MPa was artificially set, with a total force of about 1.728×10^7 N, to push the ice layer to squeeze the hull. When the supporting reaction force of the hull is about 1.728×10^7 N, the calculation stops. The calculation results at the last moment are used to verify the reliability of the ice load sensing method developed in this study.

The study simulates three different ship-ice compression load cases, including one, two, and three ice blocks compressing the ship's outer plate. To better represent the distribution of ice pressure, the zones affected by the ice is divided into multiple individual cells. It is assumed that the ice pressure is uniformly distributed within each cell. The more individual cells are divided, the closer the distribution of ice load is to reality. The numbering of individual cells in the ice load affected zone is shown in Fig. 15. It can be found from Fig. 15 that the individual cells 73 to 78, 103 to 108, 211 to 222, 325 to 330, and 355 to 360, are close to the strong frames.

Table 2 indicating the ice load conditions for the three scenarios.

4.1. Load case 1

In load case 1, the simulated total force exerted by the sea ice on the

containment segment is 1.728×10^7 N, and the resultant force on the ship's outer plate is depicted in Fig. 16.

Fig. 16 illustrates significant force fluctuations at the moment of contact between the sea ice and the ship. The structural forces tend to stabilize as the compression time increases, with the stabilized range approaching 1.72×10^7 N. The von Mises stress distribution of the FNPP hull structure after force stabilization is illustrated in Fig. 17.

The strain responses obtained from the positions optimized using BCOD, SAA, and ISAA algorithms are utilized as structural monitoring data and input into the ice load sensing mathematical model. The sensing results are depicted in Fig. 18.

The high-stress regions in the structure are located in the middle of the ice pressure zone. This is primarily due to the presence of strong frames behind the outer plate in that area, which are the main load-bearing components. Conversely, stress in the outer plate regions far from the strong frames is minimal. The sensing results of the ice load also clearly reflect this situation. Under the compression of a single ice block, the sensing results of the ice load are highly accurate. The ice load sensing effectiveness of the mathematical models optimized by the three algorithms is similar due to the absence of noise interference in the signals.

4.2. Load case 2

In load case 2, the simulated total force exerted by the sea ice on the containment segment is approximately 3.456×10^7 N, and the resultant force on the ship's outer plate is depicted in Fig. 19.

Fig. 19 illustrates significant force fluctuations at the moment of contact between the sea ice and the ship. The structural forces tend to stabilize as the compression time increases, with the stabilized range approaching 3.456×10^7 N. The von Mises stress distribution in the containment segment after force stabilization is depicted in Fig. 20.

The ice load sensing results are as follows:

The strong frames in zones A and C are located on either side of the regions, and the high-stress areas in the structure are primarily concentrated near the strong frames (see Figs. 15 and 21). The sensing results of the ice load also reflect that the pressure of the ice load on both sides of zones A and C is greater. Even under the compression of two ice blocks, the sensing results of the ice load remain highly accurate.

4.3. Load case 3

In load case 3, the simulated total force exerted by the sea ice on the containment segment is approximately 5.184×10^7 N, and the resultant force on the ship's outer plate is depicted in Fig. 22.

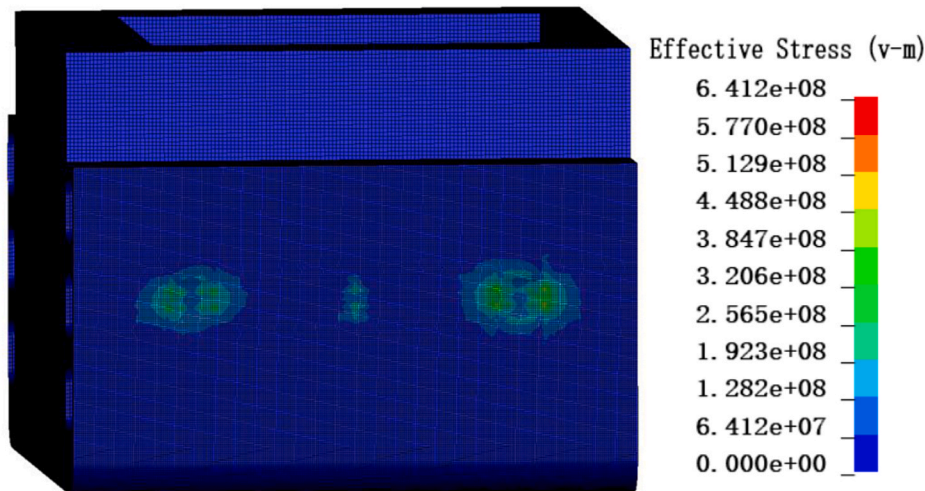


Fig. 23. Stress distribution in the containment segment.

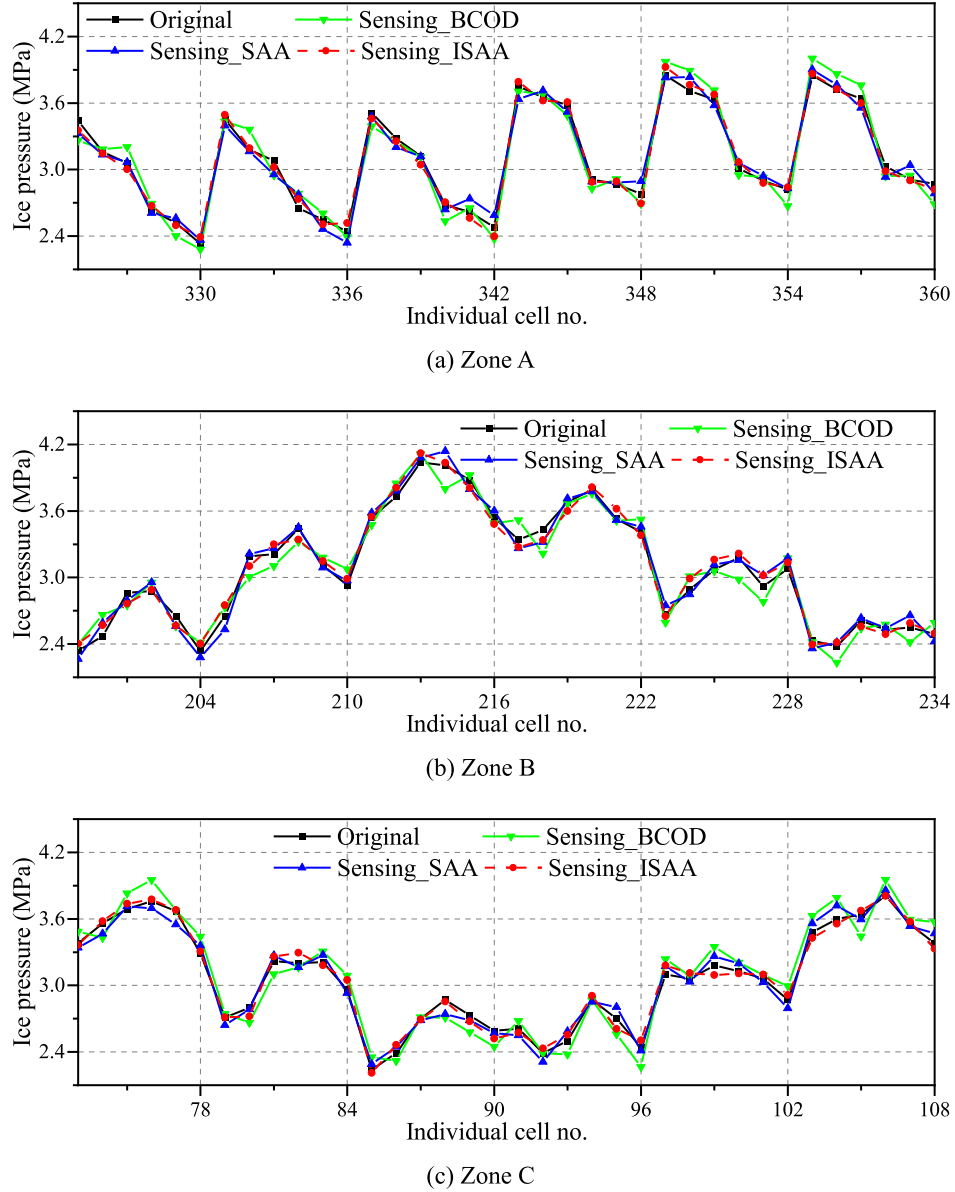


Fig. 24. Ice load sensing results (Load case 3).

Fig. 22 illustrates significant force fluctuations at the moment of contact between the sea ice and the ship. The structural forces tend to stabilize as the compression time increases, with the stabilized range approaching 5.18×10^7 N.

The von Mises stress distribution within the containment segment after force stabilization is illustrated in Fig. 23.

The ice load sensing results are as follows:

It can be found from Fig. 24 that the mathematical models corresponding to BCOD, SAA, and ISAA can provide accurate ice load sensing results, and the maximum error of the sensing results does not exceed 10%. Individual cells located near strong frame positions generally experience relatively large ice sheet compression forces, which is similar to the load cases 1 and 2. The individual cells near the strong frames are shown in Fig. 15. Furthermore, the ice load sensing results corresponding to load cases 1, 2 and 3 indicate that variations in the magnitude and distribution of the ice load compression do not affect the sensing accuracy. Additionally, the mathematical model of ice load sensing optimized by ISAA consistently exhibits the best accuracy due to its low ill posedness, but this advantage is not significant in the absence

of error interference.

5. Robustness analysis

The strain data used to sense ice loads were obtained through simulation calculations, free from noise interference. Consequently, the sensing of ice loads using mathematical models optimized by BCOD, SAA, and ISAA has shown excellent results. However, in actual environments, strain monitoring data are often subject to noise interference due to factors such as weather, temperature, and electromagnetic disturbances.

White noise signals with signal-to-noise ratios (SNR) ranging from 20 dB to 40 dB were added to the calculated strain signals to investigate the robustness of ice load sensing mathematical models optimized by different algorithms. The error levels of ice load sensing results under different noise levels were analyzed. The SNR is defined as follows:

$$SNR = 10 \times \log_{10} \left(\frac{P_{Signal}}{P_{Noise}} \right) \quad (9)$$

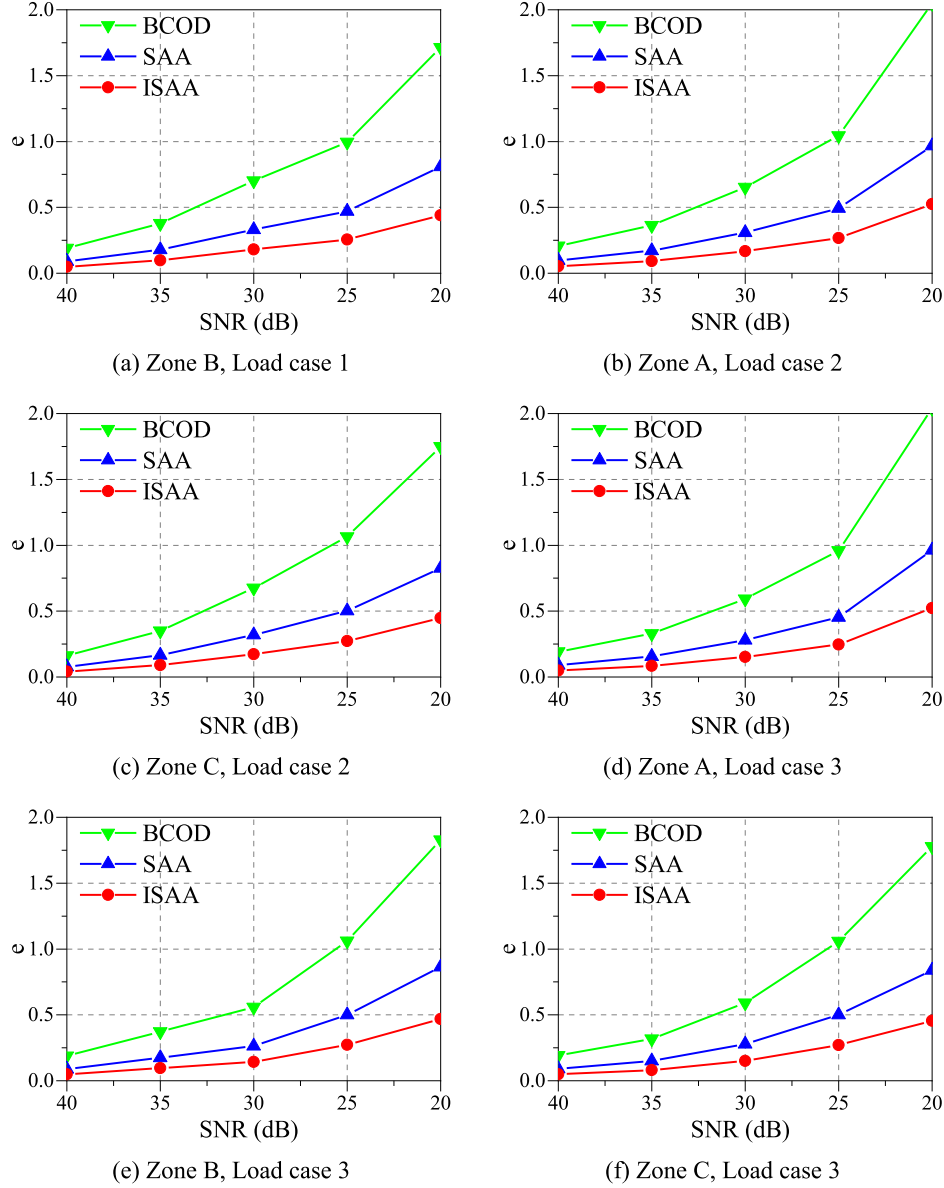


Fig. 25. The overall relative error obtained from strain signal sensing at different noise levels.

where: P_{Signal} represents the power of the signal; P_{Noise} represents the power of the noise.

The process of adding noise signals is as follows: First, the structural strain response data from 0.3s to 0.6s during the ship-ice compression process were extracted and considered to be in the stable phase of the ship-ice compression. Then, the power of the structural strain response during this period was calculated. Based on the SNR, the corresponding noise power was computed. White noise signals were generated based on the noise power and added to the structural strain response during this period. The resulting noisy strain signals were used as the actual monitoring signals, which were then input into the ice load sensing mathematical model to sense the ice loads during the stable phase.

To quantify the overall error level of the ice load sensing results within a certain area, all the sensing results on individual cells within the area were brought into the following formula to obtain the overall relative error value e :

All sensing results from individual cells within the area were incorporated into Eq. (10) to calculate the overall relative error value, thus quantifying the overall error level e of the ice load sensing results within

a specific area:

$$e = \left\{ \frac{\sum_{i=1}^n [\varepsilon(i) - \varepsilon'(i)]^2}{\sum_{i=1}^n [\varepsilon'(i)]^2} \right\}^{1/2} \quad (10)$$

where: n is the total number of individual cells; $\varepsilon(i)$ is the sensing ice load, and $\varepsilon'(i)$ is the actual ice load.

The overall relative error of ice load sensing results under strain signals with different noise levels is shown below:

The results in Fig. 25 indicate that noise significantly affects the ice load sensing results. Among them, the ice load sensing mathematical model optimized by ISAA demonstrates better robustness. In contrast, other algorithms exhibit significant sensing errors when the SNR is 30 dB. Therefore, the lower the ill-posedness of the ice load sensing mathematical model, the better the accuracy of the sensing results and the robustness of the algorithm.

6. Conclusions

The sensing of the distributed ice load on the hull plate of an icebreaker can be treated as a static inverse problem, which can be expressed in matrix form as $\mathbf{K}\mathbf{f} = \boldsymbol{\varepsilon}$. The sensing results suffer from the ill-posedness of the mathematical model. The condition number of transfer matrix $\mathbf{K}$ and the determinant of $\mathbf{K}^T\mathbf{K}$ can be used as the ill-posed index of the mathematical model $\mathbf{K}\mathbf{f} = \boldsymbol{\varepsilon}$. A low condition number and large determinant indicate accurate and stable sensing results to a certain degree.

Currently, commonly used sensor optimization methods such as DOD, COD, and BCOD, constrained by their respective search strategies and computational efficiency, fail to produce satisfactory optimization results when the number of candidate sensors exceeds 10,000, and sometimes cannot even perform computations. To address this, this study employs the SAA algorithm for optimization with a large number of candidate sensors. To improve the randomness of the SAA algorithm, a modified approach, termed ISAA, is proposed, which integrates local search and multi-start search strategies to reduce sensitivity to algorithm parameters and enhance stability. The effectiveness of the proposed algorithm is verified through the sensing of ice loads under various operating conditions. Specific conclusions are summarized as follows.

- (1) The greater the number of load-bearing elements in the ice load sensing mathematical model, the more detailed the description of the ice pressure distribution, albeit with a higher dimensionality of the mathematical model, which can significantly affect sensing results due to increased ill-posedness. Sensor optimization methods can establish mathematical models with lower ill-posedness. Under conditions of a large number of candidate sensors, SAA demonstrates higher computational efficiency and optimization effectiveness.
- (2) Although SAA exhibits high optimization efficiency, the sensitivity of optimization results to algorithm parameters and initial solutions is significant, leading to considerable randomness in the results. Hence, the proposed ISAA, integrating local search and multi-start search strategies, achieves higher stability and better optimization outcomes.
- (3) The ice compression force is simulated using Ls-Dyna. Three numerical examples demonstrate that variations in the range and magnitude of ice load do not affect the accuracy of the sensing results.
- (4) Comparative analysis of the overall relative error of ice load sensing results at different noise levels was conducted. Among them, the mathematical model for ice load sensing optimized by ISAA exhibits higher robustness. Significant sensing errors are observed with other algorithms when the SNR is 30 dB.

CRedit authorship contribution statement

Donghui Wang: Writing – original draft, Validation, Resources, Methodology, Formal analysis. **Jiajun Wang:** Software, Data curation. **Meng Zhang:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Xianqiang Qu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was sponsored and supported by the Leading Innovation Project of China National Nuclear Corporation (KY90200220002), National Natural Science Foundation of China (52301365), Natural Science Fund project of Shandong province (ZR2022QE106), Natural Science Fund project of Heilongjiang province (LH2023E075), Science and Technology Plan Project of Yantai City (2023JMRH004) and Special fund for Taishan Industrial Leading Talent Project (tssl20230605).

References

- Bazán, F.S.V., Boos, E., 2018. Schultz matrix iteration based method for stable solution of discrete ill-posed problems. *Lin. Algebra Appl.* 554, 120–145. <https://doi.org/10.1016/j.laa.2018.05.022>.
- Calvetti, D., Reichel, L., Shuibi, A., 2005. Invertible smoothing preconditioners for linear discrete ill-posed problems. *Applied Numerical Mathematics*, 6th IMACS 54, 135–149. <https://doi.org/10.1016/j.apnum.2004.09.027>.
- Comfort, G., Gong, Y., Singh, S., Abdelnour, R., 2011. Static ice loads on dams. *Can. J. Civ. Eng.* 30, 42–68. <https://doi.org/10.1139/102-080>.
- Eik, K., Marchenko, A., 2010. Model tests of iceberg towing. *Cold Reg. Sci. Technol.* 61, 13–28. <https://doi.org/10.1016/j.coldregions.2009.12.002>.
- Frederking, R., Sudom, D., 2006. Maximum ice force on the Molikpaq during the April 12, 1986 event. *Cold Reg. Sci. Technol.* 46, 147–166. <https://doi.org/10.1016/j.coldregions.2006.08.019>.
- Gagnon, R., 2008. Analysis of data from bergy bit impacts using a novel hull-mounted external Impact Panel. *Cold Regions Science and Technology*, Bergy Bits 52, 50–66. <https://doi.org/10.1016/j.coldregions.2007.04.018>.
- Gagnon, R.E., 2012. An explanation for the Molikpaq May 12, 1986 event. *Cold Reg. Sci. Technol.* 82, 75–93. <https://doi.org/10.1016/j.coldregions.2012.05.009>.
- Gong, Y., Comfort, G., Armstrong, T., Schellenberg, G., Penner, R., 1999. *Static Ice Loads on Wooden and Steel Stoplogs at Seven Sisters Generating Station*.
- Gupta, D.K., Dhinra, A.K., 2013. Input load identification from optimally placed strain gages using D-optimal design and model reduction. *Mech. Syst. Signal Process.* 40, 556–570. <https://doi.org/10.1016/j.ymssp.2013.06.011>.
- Hochstenbach, M.E., Reichel, L., 2012. Combining approximate solutions for linear discrete ill-posed problems. *Journal of Computational and Applied Mathematics*, Inverse Problems: Computation and Applications 236, 2179–2185. <https://doi.org/10.1016/j.cam.2011.09.040>.
- Jeon, M., Choi, K., Min, J.K., Ha, J.S., 2018. Estimation of local ice load by analyzing shear strain data from the IBRV ARAON's 2016 Arctic voyage. *Int. J. Nav. Archit. Ocean Eng.* 10, 421–425. <https://doi.org/10.1016/j.ijnaoe.2018.02.009>.
- Kharik, E., Roubtsova, V., Morse, B., Fafard, M., Côté, A., Taras, A., 2015. Impact of ice type on predicted ice load for dams. In: *Proceedings of the 18th Workshop on the Hydraulics of Ice Covered Rivers*, pp. 18–20. Quebec, Quebec.
- Landi, G., Loli Piccolomini, E., 2010. An iterative Lagrange method for the regularization of discrete ill-posed inverse problems. *Comput. Math. Appl.* 60, 1723–1738. <https://doi.org/10.1016/j.camwa.2010.07.003>.
- Lee, T.-K., Lee, J.-H., Kim, H., Rim, C.W., 2014. Field measurement of local ice pressures on the ARAON in the Beaufort Sea. *Int. J. Nav. Archit. Ocean Eng.* 6, 788–799. <https://doi.org/10.2478/IJNAOE-2013-0213>.
- Li, Z., Zhang, M., Ma, F., Zhu, F., Qu, X., 2024. A reconstruction method for structural stress distribution of wind turbine tower using optimised mathematical model. *Ocean. Eng.* 301, 117562. <https://doi.org/10.1016/j.oceaneng.2024.117562>.
- Liu, C.-S., 2012. Optimally scaled vector regularization method to solve ill-posed linear problems. *Appl. Math. Comput.* 218, 10602–10616. <https://doi.org/10.1016/j.amc.2012.04.022>.
- Liu, Y.H., 2017. *Research on Ice Load Inverse Method and Application of Polar Navigation Ship* (PhD Thesis). PhD Thesis]. Harbin Engineering University, Harbin.
- Loubes, J.-M., Pelletier, B., 2008. Maximum entropy solution to ill-posed inverse problems with approximately known operator. *J. Math. Anal. Appl.* 344, 260–273. <https://doi.org/10.1016/j.jmaa.2008.02.043>.
- Morigi, S., Reichel, L., Sgallari, F., Zama, F., 2006. Iterative methods for ill-posed problems and semiconvergent sequences. *J. Comput. Appl. Math.* 193, 157–167. <https://doi.org/10.1016/j.cam.2005.05.028>.
- Papadimitriou, C., 2004. Optimal sensor placement methodology for parametric identification of structural systems. *J. Sound Vib.* 278, 923–947. <https://doi.org/10.1016/j.jsv.2003.10.063>.
- Reichel, L., Sadok, H., 2008. A new l-curve for ill-posed problems. *J. Comput. Appl. Math.* 219 (2), 493–508. <https://doi.org/10.1016/j.cam.2007.01.025>.
- Reichel, L., Shyshkov, A., 2010. Cascadic multilevel methods for ill-posed problems. *Journal of Computational and Applied Mathematics*, Special Issue Dedicated to William B. Gragg on the Occasion of His 70th Birthday 233, 1314–1325. <https://doi.org/10.1016/j.cam.2009.03.019>.
- Rezghi, M., Hosseini, S.M., 2009. A new variant of L-curve for Tikhonov regularization. *J. Comput. Appl. Math.* 231, 914–924. <https://doi.org/10.1016/j.cam.2009.05.016>.
- Ritch, R., Frederking, R., Johnston, M., Browne, R., Ralph, F., 2008. Local ice pressures measured on a strain gauge panel during the CCGS Terry Fox bergy bit impact study. *Cold Regions Science and Technology*, Bergy Bits 52, 29–49. <https://doi.org/10.1016/j.coldregions.2007.04.017>.
- St John, J.W., Daley, C., Blount, H., 1990. *ICE LOADS AND SHIP RESPONSE TO ICE*.

- Su, B., Riska, K., Moan, T., 2011. Numerical study of ice-induced loads on ship hulls. *Marine Structures, MARINE STRUCTURES - AN OUTLOOK* 24, 132–152. <https://doi.org/10.1016/j.marstruc.2011.02.008>.
- Timco, G.W., Johnston, M., 2003. Ice loads on the Molikpaq in the Canadian beaufort sea. *Cold Reg. Sci. Technol.* 37, 51–86. [https://doi.org/10.1016/S0165-232X\(03\)00035-1](https://doi.org/10.1016/S0165-232X(03)00035-1).
- Wang, F., Zou, Z.-J., Zhou, L., Ren, Y.-Z., Wang, S.-Q., 2018. A simulation study on the interaction between sloping marine structure and level ice based on cohesive element model. *Cold Reg. Sci. Technol.* 149, 1–15. <https://doi.org/10.1016/j.coldregions.2018.01.022>.
- Zhang, M., Qiu, B., Kalhori, H., Qu, X., 2019a. Hybrid reconstruction method for indirect monitoring of an ice load of a steel gate in a cold region. *Cold Reg. Sci. Technol.* 162, 19–34. <https://doi.org/10.1016/j.coldregions.2019.03.020>.
- Zhang, M., Qiu, B., Zhu, M., Qu, X., 2019b. Novel computation method of reducing ill-posedness for structural static distributed load identification by optimising strain gauge locations. *Mech. Syst. Signal Process.* 124, 83–110. <https://doi.org/10.1016/j.ymssp.2019.01.051>.
- Zhang, M., Qu, X., Kalhori, H., Ye, L., 2018. Indirect monitoring of distributed ice loads on a steel gate in a cold region. *Cold Reg. Sci. Technol.* 151, 267–287. <https://doi.org/10.1016/j.coldregions.2018.04.001>.
- Zhou, L., Chuang, Z., Ji, C., 2018. Ice forces acting on towed ship in level ice with straight drift. Part I: analysis of model test data. *Int. J. Nav. Archit. Ocean Eng.* 10, 60–68. <https://doi.org/10.1016/j.ijnaoe.2017.03.008>.
- Zhou, L., Diao, F., Song, M., Han, Y., Ding, S., 2020. Calculation methods of icebreaking capability for a double-acting polar ship. *J. Mar. Sci. Eng.* 8, 179. <https://doi.org/10.3390/jmse8030179>.
- Zhou, L., Ding, S., Song, M., Gao, J., Shi, W., 2019a. A simulation of non-simultaneous ice crushing force for wind turbine towers with large slopes. *Energies* 12, 2608. <https://doi.org/10.3390/en12132608>.
- Zhou, L., Wang, F., Diao, F., Ding, S., Yu, H., Zhou, Y., 2019b. Simulation of ice-propeller collision with cohesive element method. *J. Mar. Sci. Eng.* 7, 349. <https://doi.org/10.3390/jmse7100349>.
- Zhou, Q., Peng, H., Qiu, W., 2016. Numerical investigations of ship–ice interaction and maneuvering performance in level ice. *Cold Reg. Sci. Technol.* 122, 36–49. <https://doi.org/10.1016/j.coldregions.2015.10.015>.
- Zhu, F., Zhang, M., Ma, F., Li, Z., Qu, X., 2024. Identification of wind load exerted on the jacket wind turbines from optimally placed strain gauges using C-optimal design and mathematical model reduction. *J. Mar. Sci. Eng.* 12, 563. <https://doi.org/10.3390/jmse12040563>.